

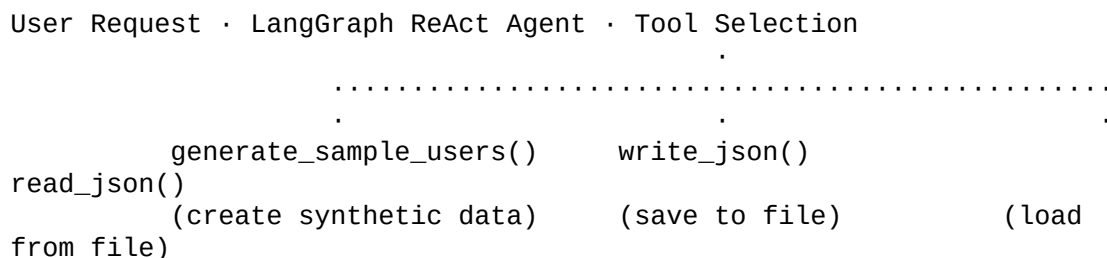
Python AI Agent in 10 Minutes

A **data generation agent** built with **LangGraph** that creates synthetic user datasets through natural language · complete with JSON persistence and conversation history.

· Features

- **·User Data Generation** · Create synthetic user datasets with customizable fields
 - **·JSON Persistence** · Read and write data to JSON files
 - **·Conversation History** · Multi-turn interactions with memory
 - **·Tool-based Agent** · LangGraph ReAct pattern with tool decorators
 - **·Quick Setup** · Minimal code, maximum capability
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.. Architecture



· Getting Started

```
cd PythonAIAgentin10Minutes
uv sync
uv run python main.py
```

Environment Variables (.env)

```
OPENAI_API_KEY=sk-...
```

· Logic Flow

- 1 **Input** · User describes what data they need (e.g. "Generate 50 users with names and emails")
- 2 **Agent Reasoning** · Agent determines the best approach
- 3 **Data Generation** · `generate_sample_users()` creates synthetic data with customizable parameters
- 4 **Persistence** · `write_json()` saves data to a JSON file
- 5 **Verification** · `read_json()` can reload and display the data
- 6 **Conversation** · History maintained for follow-up requests

Tools

Tool	Purpose	Code	Description
<code>generate_sample_users(count, fields)</code>	Create synthetic user records	<code>write_json(filename, data)</code>	Save data to JSON file
<code>read_json(filename)</code>	Load and display JSON file contents		

· Dependencies

langchain, langchain-openai, langgraph, python-dotenv

· License

Educational project · use freely for learning and reference.